

AM-GM INEQUALITY

JEE Advanced Preparation — Mathematics

AM-GM-HM Hierarchy | Proofs | Weighted AM-GM | Cauchy-Schwarz | Optimization Applications

Subject	Mathematics
Level	JEE Advanced
Topic	AM-GM Inequality — Theory, Proofs, Applications
Key Rule	$AM \geq GM \geq HM$; equality iff all quantities are equal

1.	Definitions — AM, GM, HM for n positive numbers
2.	Proof of $AM \geq GM$ (2 numbers, induction sketch)
3.	$AM \geq GM \geq HM$ chain and equality conditions
4.	Weighted AM-GM inequality
5.	Standard applications and inequalities
6.	Cauchy-Schwarz inequality
7.	Optimization techniques using AM-GM
8.	6 Fully Solved Worked Examples
9.	10 JEE Advanced Level MCQs
10.	Answer Key + Key Formulas

1. DEFINITIONS — AM, GM, HM

Arithmetic Mean (AM) of $a_1, a_2, \dots, a_n$ (all positive):

$$AM = (a_1 + a_2 + \dots + a_n) / n$$

Geometric Mean (GM):

$$GM = (a_1 * a_2 * \dots * a_n)^{1/n} \text{ [n-th root of the product]}$$

Harmonic Mean (HM):

$$HM = n / (1/a_1 + 1/a_2 + \dots + 1/a_n) \text{ [reciprocal of avg of reciprocals]}$$

THE FUNDAMENTAL INEQUALITY:

$$AM \geq GM \geq HM$$

Equality holds in each case if and only if ALL quantities are EQUAL.

2. PROOF OF $AM \geq GM$

2.1 Proof for 2 Numbers (Direct Algebraic)

Claim: For positive a, b : $(a+b)/2 \geq \sqrt{ab}$

Proof:

$$(a+b)/2 \geq \sqrt{ab}$$

$$\Leftrightarrow a+b \geq 2\sqrt{ab} \text{ [multiply both sides by 2]}$$

$$\Leftrightarrow a + b - 2\sqrt{ab} \geq 0$$

$$\Leftrightarrow (\sqrt{a} - \sqrt{b})^2 \geq 0 \text{ [which is always true!]}$$

Equality holds iff $\sqrt{a} = \sqrt{b}$, i.e., $a = b$.

Equivalently: $a + b \geq 2\sqrt{ab}$ is just $(\sqrt{a} - \sqrt{b})^2 \geq 0$.

2.2 Proof for 4 Numbers (Bootstrap Method)

Given 2-variable AM-GM, prove for 4 variables:

$$(a+b+c+d)/4 \geq [(a+b)/2 * (c+d)/2]^{1/2} \geq [\sqrt{ab} * \sqrt{cd}]^{1/2} = (abcd)^{1/4}$$

Steps: apply 2-var AM-GM to pairs (a,b) , (c,d) , then to their means.

This 'doubling' technique extends AM-GM to $n=2^k$ numbers.

For general n : use Cauchy's forward-backward induction.

2.3 Proof of $GM \geq HM$ for 2 Numbers

$$\text{Need: } \sqrt{ab} \geq \frac{2ab}{a+b}$$

$$\Leftrightarrow \sqrt{ab} * (a+b) \geq 2ab$$

$$\Leftrightarrow (a+b) \geq 2\sqrt{ab} \text{ [divide both sides by } \sqrt{ab}]$$

This is just AM $\geq$ GM again! So GM $\geq$ HM follows from AM $\geq$ GM.

3. WEIGHTED AM-GM INEQUALITY

Let $a_1, \dots, a_n > 0$ and weights $w_1, \dots, w_n > 0$ with $w_1 + \dots + w_n = 1$.

Then: $w_1 a_1 + w_2 a_2 + \dots + w_n a_n \geq a_1^{w_1} a_2^{w_2} \dots a_n^{w_n}$

Equality iff $a_1 = a_2 = \dots = a_n$.

Important special case (equal weights $w_i = 1/n$):

$$(a_1 + \dots + a_n)/n \geq (a_1 \dots a_n)^{1/n} \text{ [ordinary AM-GM]}$$

Weighted AM-GM for maximizing $x^a * y^b * z^c$ ($a+b+c=1$):

Given $x+y+z=1$: by weighted AM-GM with $w_1=a, w_2=b, w_3=c$:

$$ax/a + by/b + cz/c \geq (x/a)^a * (y/b)^b * (z/c)^c$$

i.e., $x+y+z \geq (x/a)^a (y/b)^b (z/c)^c \Rightarrow$ product maximized when $x=a, y=b, z=c$

4. STANDARD INEQUALITIES DERIVED FROM AM-GM

Inequality 1: $a^2 + b^2 \geq 2ab$ for all real a, b

Proof: $a^2 + b^2 - 2ab = (a-b)^2 \geq 0$. Equality iff $a=b$.

Inequality 2: $a^2 + b^2 + c^2 \geq ab + bc + ca$ for positive a, b, c

Proof: Multiply both sides by 2:

$$2(a^2 + b^2 + c^2) \geq 2(ab + bc + ca)$$

$$\Leftrightarrow (a^2 - 2ab + b^2) + (b^2 - 2bc + c^2) + (c^2 - 2ca + a^2) \geq 0$$

$$\Leftrightarrow (a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0 \text{ [always true!]}$$

Inequality 3: $(x + 1/x) \geq 2$ for $x > 0$

Proof: By AM-GM: $(x + 1/x)/2 \geq \sqrt{x * 1/x} = \sqrt{1} = 1$

So $x + 1/x \geq 2$. Equality when $x = 1/x$, i.e., $x = 1$.

Inequality 4: $a/b + b/c + c/a \geq 3$ for positive a, b, c

By AM-GM: $(a/b + b/c + c/a)/3 \geq (a/b * b/c * c/a)^{1/3} = 1$

So $a/b + b/c + c/a \geq 3$. Equality when $a=b=c$.

5. CAUCHY-SCHWARZ INEQUALITY

Statement:

$$(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \geq (a_1b_1 + a_2b_2 + \dots + a_nb_n)^2$$

Equality iff $a_1/b_1 = a_2/b_2 = \dots = a_n/b_n$ (all ratios equal).

Proof for 2 variables:

$$(a_1^2 + a_2^2)(b_1^2 + b_2^2) - (a_1b_1 + a_2b_2)^2$$

$$= a_1^2b_2^2 + a_2^2b_1^2 - 2a_1b_1a_2b_2$$

$$= (a_1b_2 - a_2b_1)^2 \geq 0 \text{ [always!]}$$

JEE Application: Find max of $px+qy$ subject to $x^2+y^2=1$:

By Cauchy-Schwarz: $(p^2+q^2)(x^2+y^2) \geq (px+qy)^2$

$$\Rightarrow (px+qy)^2 \leq p^2+q^2 \Rightarrow \max(px+qy) = \sqrt{p^2+q^2}$$

Achieved when $x/p = y/q$, i.e., $x=p/\sqrt{p^2+q^2}$, $y=q/\sqrt{p^2+q^2}$.

6. OPTIMIZATION USING AM-GM

Strategy for finding minimum/maximum using AM-GM:

1. Identify the expression to minimize/maximize.
2. Split the expression into parts whose product is CONSTANT.
3. Apply $AM \geq GM$; the minimum of the sum equals $n * (\text{constant product})^{(1/n)}$.
4. Find when equality holds (all parts equal).

Example: Minimize $f(x,y) = x^2+y^2$ given $xy=1$.

By AM-GM: $(x^2+y^2)/2 \geq \sqrt{x^2y^2} = \sqrt{(xy)^2} = 1$

So $x^2+y^2 \geq 2$. Equality when $x^2=y^2$ and $xy=1$, i.e., $x=y=1$.

Example: Maximum of abc given $a+b+c = S$ (fixed).

By AM-GM: $(a+b+c)/3 \geq (abc)^{(1/3)}$

So $abc \leq (S/3)^3$. Equality when $a=b=c=S/3$.

7. WORKED EXAMPLES

Example 1: Find the minimum value of $f(x) = x + 4/x$ for $x > 0$

Solution:

Apply AM-GM to the two terms x and $4/x$:

$$(x + 4/x)/2 \geq \sqrt{x \cdot 4/x} = \sqrt{4} = 2$$

$$\text{So } x + 4/x \geq 4$$

Equality when $x = 4/x$, i.e., $x^2 = 4$, i.e., $x = 2$ ($x > 0$)

Verify: $f(2) = 2 + 4/2 = 2 + 2 = 4$ [confirmed!]

ANSWER: Minimum value = 4, achieved at $x = 2$

Example 2: Find the minimum of $(x+y)(1/x + 1/y)$ for $x, y > 0$

Solution:

$$(x+y)(1/x+1/y) = 1 + x/y + y/x + 1 = 2 + x/y + y/x$$

By AM-GM: $x/y + y/x \geq 2\sqrt{(x/y) \cdot (y/x)} = 2\sqrt{1} = 2$

$$\text{So } (x+y)(1/x+1/y) \geq 2 + 2 = 4$$

Equality when $x/y = y/x$, i.e., $x = y$

ANSWER: Minimum = 4, achieved when $x = y$

Example 3: Prove: For positive a, b, c : $a/b + b/c + c/a \geq 3$

Solution:

$$\text{Let } p = a/b, q = b/c, r = c/a$$

$$\text{Note: } p \cdot q \cdot r = (a/b) \cdot (b/c) \cdot (c/a) = 1$$

$$\text{By AM-GM: } (p+q+r)/3 \geq (p \cdot q \cdot r)^{1/3} = 1^{1/3} = 1$$

$$\text{So } p + q + r \geq 3$$

$$\text{i.e., } a/b + b/c + c/a \geq 3$$

Equality when $p=q=r=1$, i.e., $a=b=c$.

ANSWER: Proved. Equality when $a = b = c$.

Example 4: Find max value of $2x+3y$ given $x^2+y^2=1$ (use Cauchy-Schwarz)

Solution:

$$\text{By Cauchy-Schwarz: } (2^2+3^2)(x^2+y^2) \geq (2x+3y)^2$$

$$(4+9)(1) \geq (2x+3y)^2$$

$$13 \geq (2x+3y)^2$$

$$\text{So } 2x+3y \leq \sqrt{13}$$

Equality when $x/2 = y/3$, combined with $x^2+y^2=1$:

Let $x=2k$, $y=3k \Rightarrow 4k^2+9k^2=1 \Rightarrow k=1/\sqrt{13}$

At this point: $2x+3y = 2*(2/\sqrt{13}) + 3*(3/\sqrt{13}) = 13/\sqrt{13} = \sqrt{13}$ [confirmed!]

ANSWER: Maximum = $\sqrt{13}$, achieved at $x=2/\sqrt{13}$, $y=3/\sqrt{13}$

Example 5: If $a+b+c=1$ with $a,b,c>0$, find the maximum of $a*b*c$

Solution:

By AM-GM: $(a+b+c)/3 \geq (abc)^{1/3}$

$1/3 \geq (abc)^{1/3}$

$(1/3)^3 \geq abc$

$abc \leq 1/27$

Equality when $a = b = c = 1/3$

ANSWER: Maximum $abc = 1/27$, achieved when $a = b = c = 1/3$

Example 6: Prove: For n positive numbers $a_1, \dots, a_n$, their AM $\geq$ GM

Solution:

Use Cauchy's forward-backward induction:

Base: $n=2$: $(a+b)/2 \geq \sqrt{ab}$ proved by $(a-b)^2 \geq 0$.

Forward step: if true for $n=k$, prove for $n=2k$.

$AM(a_1, \dots, a_{2k}) = [AM(a_1 \dots a_k) + AM(a_{k+1} \dots a_{2k})] / 2$

$\geq [GM(a_1 \dots a_k) * GM(a_{k+1} \dots a_{2k})]^{1/2}$ [by 2-var AM-GM]

$= [a_1 \dots a_{2k}]^{1/(2k)} = GM(a_1, \dots, a_{2k})$

Backward step: if true for $n=k$, prove for $n=k-1$.

Set $a_k = AM(a_1, \dots, a_{k-1}) = A$. Then AM of k numbers = A .

By hypothesis: $A \geq (a_1 * \dots * a_{k-1} * A)^{1/k}$

$A^k \geq a_1 * \dots * a_{k-1} * A \Rightarrow A^{k-1} \geq a_1 * \dots * a_{k-1}$

$A \geq (a_1 * \dots * a_{k-1})^{1/(k-1)} = GM(a_1, \dots, a_{k-1})$. QED.

ANSWER: AM $\geq$ GM for all n positive numbers by induction.

8. PRACTICE QUESTIONS — JEE ADVANCED LEVEL

Q1. [Single Correct] The minimum value of $x + 9/x$ for $x > 0$ is:

- (A) 3
- (B) 6
- (C) 9
- (D) 12

Q2. [Single Correct] For positive a, b, c with $a+b+c=6$, the maximum value of abc is:

- (A) 4
- (B) 6
- (C) 8
- (D) 27

Q3. [Single Correct] If AM of two positive numbers is 10 and GM is 8, then HM is:

- (A) 5.4
- (B) 6.4
- (C) 6.0
- (D) 7.2

Q4. [Multiple Correct] Which of the following are always true for positive a, b ?

- (A) $a+b \geq 2\sqrt{ab}$
- (B) $a^2+b^2 \geq 2ab$
- (C) $(a+b)/2 \geq \sqrt{ab} \geq 2ab/(a+b)$
- (D) $\sqrt{ab} \geq (a+b)/2$

Q5. [Single Correct] Maximum of $(\sin(x))^2 * \cos(x)$ for x in $(0, \pi/2)$ using AM-GM:

- (A) $2\sqrt{3}/9$
- (B) $\sqrt{3}/3$
- (C) $2/(3\sqrt{3})$
- (D) $1/3$

Q6. [Single Correct] By Cauchy-Schwarz, max of $3x+4y$ given $x^2+y^2=1$ is:

- (A) 4
- (B) 5
- (C) 7
- (D) 12

Q7. [Single Correct] For positive x, y, z with $xyz=8$, the minimum of $x+y+z$ is:

- (A) 2
- (B) 4
- (C) 6
- (D) 8

Q8. [Multiple Correct] For $a+b+c=1$ ($a, b, c > 0$):

- (A) $abc \leq 1/27$
- (B) $ab+bc+ca \leq 1/3$
- (C) $a^2+b^2+c^2 \geq 1/3$
- (D) $(1/a)+(1/b)+(1/c) \geq 9$

Q9. [Single Correct] The minimum value of $(x+y)(1/x+4/y)$ for $x, y > 0$ is:

- (A) 4

- (B) 6
- (C) 9
- (D) 10

Q10. [Single Correct] Minimum of $a/b + b/c + c/a$ for positive a, b, c is:

- (A) 1
- (B) 2
- (C) 3
- (D) Cannot be determined

9. ANSWER KEY WITH SOLUTIONS

Q#	Answer	Key Solution
Q1	(B) 6	$x+9/x \geq 2\sqrt{9} = 6$; equality at $x=3$
Q2	(C) 8	AM-GM: $(a+b+c)/3 \geq \sqrt[3]{abc}$; $\max abc = (6/3)^3 = 2^3 = 8$; at $a=b=c=2$
Q3	(B) 6.4	HM = $2ab/(a+b)$; find a, b : $a+b=20, ab=64$; HM = $2 \cdot 64 / 20 = 6.4$
Q4	A, B, C	(D) reverses the inequality — false
Q5	(C) $2/(3\sqrt[3]{3})$	Write as $\sin^2(x) \cos(x)$; use AM-GM on 3 terms
Q6	(B) 5	CS: $(9+16)(x^2+y^2) \geq (3x+4y)^2$; $\max = \sqrt{25} = 5$
Q7	(C) 6	AM-GM: $(x+y+z)/3 \geq \sqrt[3]{xyz} = 2$; $\min x+y+z=6$ at $x=y=z=2$
Q8	A, B, C, D	All follow from AM-GM applied to $a+b+c=1$
Q9	(C) 9	$(x+y)(1/x+4/y) = 1+4x/y+y/x+4$; AM-GM on $4x/y+y/x \geq 4$; total ≥ 9
Q10	(C) 3	$pqr=1 \Rightarrow$ AM-GM gives $\text{sum} \geq 3$

10. KEY FORMULAS — QUICK REFERENCE

Concept	Formula / Statement
AM-GM (2 vars)	$(a+b)/2 \geq \sqrt{ab}$; equality iff $a=b$
AM-GM (n vars)	$(a_1+\dots+a_n)/n \geq (a_1 \dots a_n)^{1/n}$
GM $\geq$ HM	$\sqrt{ab} \geq 2ab/(a+b)$
Chain	AM $\geq$ GM $\geq$ HM; equality iff all equal
$x + c/x$ (min)	$\min = 2\sqrt{c}$ at $x=\sqrt{c}$
$a+b+c = S$ (max abc)	$\max abc = (S/3)^3$ at $a=b=c=S/3$
Cauchy-Schwarz	$(\sum a_i^2)(\sum b_i^2) \geq (\sum a_i b_i)^2$
Max of $px+qy$ ($x^2+y^2=1$)	$\sqrt{p^2+q^2}$
$a^2+b^2+c^2 \geq ab+bc+ca$	Always true; from $(a-b)^2+(b-c)^2+(c-a)^2 \geq 0$
$a/b+b/c+c/a$	≥ 3 ; equality iff $a=b=c$

ADVANCED NOTES: AM-GM Inequality

Advanced AM-GM Techniques for JEE

Technique 1 (Splitting): To minimize $ax + b/x^2$, write $ax = ax/2 + ax/2$ (3 terms).

$$ax/2 + ax/2 + b/x^2 \geq 3 \cdot (ax/2 \cdot ax/2 \cdot b/x^2)^{1/3} = 3 \cdot (a^2 b/4)^{1/3}.$$

Technique 2 (Substitution): Let $x = \tan(t)$, use sin and cos for constrained optimization.

Technique 3 (SOS — Sum of Squares): Write inequality as sum of (something)² ≥ 0 .

Technique 4: Use AM-GM on cleverly chosen k terms where product = constant.

Schur's Inequality (Extension)

Schur's inequality: For non-negative a, b, c and $t > 0$:

$$a^t(a-b)(a-c) + b^t(b-a)(b-c) + c^t(c-a)(c-b) \geq 0$$

For $t=1$: $a^3 + b^3 + c^3 + abc \geq ab(a+b) + bc(b+c) + ca(c+a)$.

This is occasionally tested in JEE Advanced olympiad-style questions.

For JEE, the cases $t=1$ and $t=2$ are most relevant.

JEE ADVANCED EXAM STRATEGY

Recognise the Type First: Before computing, classify whether it is AP/GP/AGP/Special series.

Use Standard Results: Memorise S_1 , S_2 , S_3 , and the product series formulas — saves huge time.

Telescoping Check: If T_r has product of consecutive integers, try partial fractions first.

AM-GM Shortcut: For min/max with a constraint, always try AM-GM before calculus.

Verification: Always verify with $n=1$ or $n=2$ before writing final answer.

AGP Infinite Sum: If $|r| < 1$, use $S_{\infty} = a/(1-r) + dr/(1-r)^2$ directly.

Nicomachus: Any time S_3 appears, immediately replace with $(S_1)^2$.

Multiple Correct MCQs: All four options may be true — verify each independently.

COMPLETE FORMULA CONSOLIDATION

#	Series / Expression	Closed Form
1	Sum r ($r=1$ to n)	$n(n+1)/2$
2	Sum r^2 ($r=1$ to n)	$n(n+1)(2n+1)/6$
3	Sum r^3 ($r=1$ to n)	$[n(n+1)/2]^2$
4	Sum $r(r+1)$	$n(n+1)(n+2)/3$
5	Sum $r(r+1)(r+2)$	$n(n+1)(n+2)(n+3)/4$

6	Sum $r \cdot r!$ ($r=1$ to n)	$(n+1)! - 1$
7	Sum odd numbers (n terms)	n^2
8	Sum $(2r-1)^2$ ($r=1$ to n)	$n(2n-1)(2n+1)/3$
9	Sum $(2r)^2$ ($r=1$ to n)	$2n(n+1)(2n+1)/3$
10	Sum $(2r-1)^3$ ($r=1$ to n)	$n^2(2n^2-1)$
11	$1/(r(r+1))$ — telescoping	Sum to $n = n/(n+1)$
12	$r/(r+1)!$ — telescoping	Sum to $n = 1 - 1/(n+1)!$
13	AP n th term	$a + (n-1)d$
14	AP sum n terms	$n[2a + (n-1)d]/2$
15	GP n th term	$a \cdot r^{(n-1)}$
16	GP sum infinity	$a/(1-r)$ for $ r < 1$
17	AGP sum infinity	$a/(1-r) + dr/(1-r)^2$
18	$1+2x+3x^2+\dots$ ($ x < 1$)	$1/(1-x)^2$
19	Sum $r/2^r$ ($r=1$ to ∞)	2
20	AM $\geq$ GM $\geq$ HM	Always; equality iff all equal